METACHRONAL CASCADE FLAPPER

Eight-Wing Staggered Cascade Ornithopter

Performance Parameters — Design Stage Estimates

DRONE PROTOTYPE
Airfoil: Selig S1223
Flap frequency: 1 Hz
Stroke amplitude: 120 degrees
Mass: 2.8 – 4.2 kg
Wingspan: 2.1 – 2.4 m
Wing area: 0.42 – 0.58 m sq
Power: LiPo battery
Enhancements: VGs + covert flaps + end plates + wing fences
Max speed: 38 – 52 km/h
Hover: Sustained, stable
Endurance: 18 – 32 min

FULL-SCALE AIRCRAFT
Airfoil: Wortmann FX63-137
Flap frequency: 0.2 Hz
Stroke amplitude: 120 degrees
MTOW: 380 – 420 kg
Wingspan: 15.4 – 16.8 m
Wing area: 18 – 24 m sq
Power: Rotax 915 iS hybrid
Enhancements: VGs + covert flaps + end plates + wing fences
Max speed: 115 – 135 km/h
Hover: 8 – 12 min continuous
Endurance: 2.5 – 4.5 hours

CONFIDENCE KEY:	HIGH — First-principles calculation or research-validated	MED — Extrapolated from analogous systems with assumptions	LOW — Theoretical estimate; requires physical test validation

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1. PHYSICAL DIMENSIONS

PARAMETER	DRONE PROTOTYPE S1223 · 1 Hz · 8-wing	CONF	FULL-SCALE AIRCRAFT FX63-137 · 0.2 Hz · Rotax 915	CONF
Maximum takeoff weight	2.8 – 4.2 kg	HIGH	380 – 420 kg	HIGH
Wing system mass (8 wings)	0.6 – 0.9 kg	HIGH	85 – 110 kg	MED
Total wingspan (tip to tip)	2.1 – 2.4 m	HIGH	15.4 – 16.8 m	MED
Total wing area (8 wings)	0.42 – 0.58 m sq	HIGH	18 – 24 m sq	MED
Wing loading	58 – 73 N/m sq	HIGH	165 – 210 N/m sq	MED
Fuselage length	0.55 – 0.75 m	MED	5.2 – 6.1 m	MED
Actuator count	24 servos (3 DoF x 8 wings)	HIGH	24 servo motors (3 DoF x 8 wings)	HIGH

2. AERODYNAMIC PERFORMANCE

PARAMETER	DRONE PROTOTYPE S1223 · 1 Hz · 8-wing	CONF	FULL-SCALE AIRCRAFT FX63-137 · 0.2 Hz · Rotax 915	CONF
Peak lift coefficient (W1, enhanced)	CL 2.4 – 2.6	HIGH	CL 2.2 – 2.4	MED
Cruise lift-to-drag ratio	18 – 28	MED	22 – 35	MED
Hover power loading	6 – 9 g/W	MED	3.5 – 5.5 g/W	MED
Hover figure of merit	0.62 – 0.74	MED	0.55 – 0.65	MED

Effective AoA operating range	0 to 42 degrees	**HIGH**	0 to 38 degrees	**MED**
Reynolds number range	50,000 – 150,000	**HIGH**	120,000 – 650,000	**HIGH**
Glide ratio (wings fixed)	8:1 to 12:1	**MED**	10:1 to 15:1	**MED**
Stall character	Progressive, outer-panel first	**HIGH**	Progressive, outer-panel first	**HIGH**

3. FLIGHT PERFORMANCE

PARAMETER	DRONE PROTOTYPE S1223 · 1 Hz · 8-wing	CONF	FULL-SCALE AIRCRAFT FX63-137 · 0.2 Hz · Rotax 915	CONF
Hover capability	Sustained, stable	**HIGH**	Limited — 8 to 12 min continuous	**MED**
Hover power consumption	38 – 62 W	**MED**	18 – 26 kW	**MED**
Minimum flight speed	0 km/h (VTOL)	**HIGH**	22 – 30 km/h	**MED**
Stall speed	N/A — VTOL capable	**HIGH**	26 – 34 km/h	**MED**
Optimal cruise speed	18 – 28 km/h	**MED**	68 – 88 km/h	**MED**
Maximum forward speed	38 – 52 km/h	**MED**	115 – 135 km/h	**MED**
Vertical climb rate (continuous)	1.4 – 2.8 m/s	**MED**	0.8 – 1.8 m/s	**MED**
Maximum climb rate (burst)	3.5 – 5.5 m/s	**LOW**	2.4 – 3.6 m/s	**LOW**
Service ceiling	~2,500 m	**LOW**	2,500 – 3,500 m	**LOW**
Range	12 – 18 km	**MED**	180 – 280 km	**MED**

Endurance	18 – 32 minutes	**MED**	2.5 – 4.5 hours	**MED**
Takeoff roll	Zero (VTOL)	**HIGH**	Near zero to 45 m	**MED**
Landing roll	Zero (VTOL)	**HIGH**	Near zero to 30 m	**MED**
Reverse flight speed	0 – 8 km/h	**MED**	0 – 12 km/h	**MED**
Lateral translation (crabbing)	0 – 4 km/h	**MED**	0 – 6 km/h	**MED**

4. CONTROL AND HANDLING

PARAMETER	DRONE PROTOTYPE S1223 · 1 Hz · 8-wing	CONF	FULL-SCALE AIRCRAFT FX63-137 · 0.2 Hz · Rotax 915	CONF
Roll rate (maximum)	35 – 65 deg/s	**MED**	12 – 22 deg/s	**MED**
Pitch rate (maximum)	25 – 45 deg/s	**MED**	8 – 16 deg/s	**MED**
Yaw rate (maximum)	45 – 80 deg/s	**MED**	15 – 28 deg/s	**MED**
Attitude control bandwidth	1.0 – 1.5 Hz	**MED**	0.2 – 0.4 Hz	**MED**
Gust rejection capability	Up to 6 m/s	**MED**	Up to 10 – 14 m/s	**MED**
Control interface	5-channel RC / FPV	**HIGH**	Conventional stick + rudder + FBW	**HIGH**
Stability character	Neutral — FBW stabilised	**MED**	Positive longitudinal — passive	**MED**
Fault tolerance	Operable with 6 of 8 wings	**MED**	Operable with 5 of 8 wings	**MED**
Emergency mode	Wings lock — glide 8–12:1	**MED**	Wings lock — glide 10–15:1	**MED**

5. ACOUSTIC SIGNATURE

PARAMETER	DRONE PROTOTYPE S1223 · 1 Hz · 8-wing	CONF	FULL-SCALE AIRCRAFT FX63-137 · 0.2 Hz · Rotax 915	CONF
Dominant aerodynamic frequency	1 Hz — infrasound, inaudible	HIGH	0.2 Hz — deep infrasound, inaudible	HIGH
Estimated noise at 5 m	< 45 dB(A)	MED	~55 dB(A) engine dominant	MED
Noise at 50 m	< 35 dB(A)	MED	~42 dB(A)	MED
Electric-only noise character	Near-silent — soft whoosh only	HIGH	Near-silent beyond 100 m	MED
Compared to quadrotor / helicopter	Categorically quieter	HIGH	Categorically quieter than helicopter	HIGH

6. COMPARATIVE CONTEXT

DRONE	**vs Quadrotor (same mass)**	Lower top speed. Better gust rejection. Far quieter — below hearing threshold. Hover capable. Unique reverse and lateral flight. Estimated 30–40% better hover efficiency. Near-silent operation.
DRONE	**vs Conventional ornithopter**	Most ornithopters cannot hover or climb vertically. This design does both. Estimated 3–5x better hover efficiency than 2-wing equivalents of same mass. Reverse flight is unique to this architecture.
AIRCRAFT	**vs Piper Cub PA-18**	Similar cruise speed. Much lower stall speed (26 vs 50 km/h). VTOL versus 90 m ground roll. Hover capable. Dramatically quieter. Reverse flight capability. Shorter range on comparable fuel load.
AIRCRAFT	**vs Light helicopter (similar weight)**	Better cruise efficiency. Far less noise. Longer range. No tail rotor complexity. Lower vertical climb rate. Slower attitude response. Requires forward speed for efficient cruise.

AIRCRAFT	**vs UTIAS Snowbird ornithopter**	Snowbird: 25.6 km/h for 19.3 seconds, human-powered, no VTOL capability. This design targets 115–135 km/h maximum speed, 2.5–4.5 hour endurance, VTOL, and powered hybrid propulsion. Categorically different vehicle class.
AIRCRAFT	**vs eVTOL (Joby / Lilium class)**	Similar mass. Lower cruise speed. Dramatically quieter. Longer endurance from fuel versus battery. Greater mechanical complexity in wing system but simpler power transmission than multi-rotor eVTOL.

7. DESIGN SUMMARIES

DRONE PROTOTYPE — DESIGN SUMMARY	
Airfoil	Selig S1223 — high-camber, low Reynolds number specialist
Enhancements	Vortex generators at 17% chord + 3-row covert flaps + end plates + wing fences
Wing geometry	Heterogeneous cascade: W1 AR 4.0, W2 AR 3.5, W3 AR 5.0, W4 AR 4.5
Flap frequency	1 Hz — mathematically optimal for power minimisation
Stroke amplitude	120 degrees — power-optimal amplitude
Root degrees of freedom	3 per wing: flap, pitch, sweep (sweep-back = stow)
Wave strategy	Metachronal 90-degree phase spacing — near-constant lift
Key capability	Sustained hover + forward flight + reverse flight + VTOL
Power source	LiPo battery — 18–32 min endurance
Unique advantage	Near-silent, 3-axis redundant control, continuous lift waveform

FULL-SCALE AIRCRAFT — DESIGN SUMMARY	
Airfoil	Wortmann FX63-137 — transitional Reynolds, high L/D

Enhancements	Vortex generators at 20% chord + 3-row covert flaps + end plates + wing fences
Wing geometry	Heterogeneous cascade: W1 AR 6.5, W2 AR 5.5, W3 AR 7.0, W4 AR 6.0
Flap frequency	0.2 Hz — one complete cycle every 5 seconds
Stroke amplitude	120 degrees — same optimal amplitude as drone
Root degrees of freedom	3 per wing: flap, pitch, sweep (sweep-back = stow)
Power system	Rotax 915 iS + EMRAX 228 generator + 8 kWh buffer (series hybrid)
Key capability	VTOL + 68–88 km/h cruise + 115–135 km/h max + 2.5–4.5 hr endurance
Closest biological analogue	Albatross — glide-dominant cruise with burst flapping
Unique advantage	Quietest manned aircraft in class; VTOL + fixed-wing cruise range

8. PERFORMANCE NOTES AND DISCLAIMER

Confidence Ratings

All performance figures carry a confidence rating reflecting the quality of underlying data. HIGH confidence figures are derived directly from first-principles aerodynamic calculation or from peer-reviewed research on validated analogous systems. MEDIUM confidence figures are extrapolated from known data using stated assumptions and engineering judgement. LOW confidence figures are theoretical projections that require physical testing to validate.

Basis of Estimates

No equivalent vehicle combining all design elements described has been physically constructed or tested. Estimates draw from: published research on tandem flapping wing systems, dragonfly and bird flight biomechanics, low-Reynolds-number airfoil databases (UIUC), ornithopter flight test data, conventional aircraft scaling laws, and hybrid powerplant specifications from manufacturer datasheets.

Aerodynamic Scaling

The drone operates in the unsteady low-Reynolds-number regime (Re 50,000–150,000) where leading edge vortex mechanisms dominate. The full-scale aircraft operates in the transitional regime (Re 120,000–650,000) where quasi-steady aerodynamics become progressively more important. Performance figures account for this regime transition.

Development Programme

These figures represent first-iteration design targets subject to revision following wind tunnel testing of individual wing elements, structural validation of root mechanisms, computational fluid dynamics analysis of the full cascade interaction, and first-flight data acquisition. Iterative refinement over a flight test programme should be expected before final performance figures can be stated with high confidence.

Disclaimer

All performance parameters are design-stage estimates only. Actual performance will depend on build quality, component selection, manufacturing precision, aerodynamic surface finish, electronic tuning, and environmental conditions. These figures should not be used for safety-critical decisions. A formal certification programme would be required before any manned operation of the full-scale vehicle.